

Experiment No. 5

1. Experiment Title

Apply Array Methods and Object Handling; Create a Cart Total Calculator with Discount Logic

2. Software/Tools Required

- Visual Studio Code
 - Google Chrome / Microsoft Edge
 - HTML5
 - CSS3
 - JavaScript (ES6)
-

3. Theory

Array in JavaScript

An array is a special variable that can store multiple values in a single variable. Arrays are useful for storing collections of related data.

Example:

```
let products = ["Rice", "Oil", "Sugar"];
```

Object in JavaScript

An object stores data in the form of key-value pairs. It is used to represent real-world entities such as products, customers, and employees.

Example:

```
const product = {  
  name: "Rice",  
  price: 80,  
  quantity: 5  
}
```

```
};
```

Array Methods Used

1. push()

Adds a new element to the end of an array.

```
cart.push(item);
```

2. forEach()

Executes a function for each element in an array.

```
cart.forEach(item => {  
  console.log(item);  
});
```

3. map()

Creates a new array by transforming existing elements.

```
let totals = cart.map(item => item.total);
```

4. reduce()

Reduces all array elements into a single value.

```
let grandTotal = totals.reduce((sum, value) => sum + value, 0);
```

5. filter()

Creates a new array containing elements that satisfy a condition.

```
let expensiveItems = cart.filter(item => item.price > 1000);
```

4. Output:

 **Shopping Cart Calculator**

Add To Cart

Product	Price	Qty	Total
Apple	₹40	2	₹80
Sofit	₹37	5	₹185
Notebook	₹80	6	₹480
Biscuits	₹40	4	₹160
Lays Chips	₹20	10	₹200
Rice	₹110	5	₹550
Cooking Oil	₹180	3	₹540
Dry Fruits	₹1200	2	₹2400
Bag	₹1700	1	₹1700

Calculate Final Bill

Billing Summary
Total Products : 9
Grand Total : ₹6295.00
Discount : 20%
Discount Amount : ₹1259.00
Final Bill : ₹5036.00
Expensive Products (> ₹1000): 2

5. Result/Conclusion

The experiment was successfully implemented using **HTML5, CSS3, and JavaScript**. Various array methods such as **push()**, **forEach()**, **map()**, **reduce()**, and **filter()** were used to manage and process cart data efficiently. Product details were stored using JavaScript objects, and discount logic was applied based on the total purchase amount. The Shopping Cart Total Calculator accurately generated the final bill and demonstrated the practical application of arrays, objects, and array methods in web development.